HeadGenome: Dynamic Entropy-Collapse Probing Does Not Identify Causal Attention Circuits
A Multi-Architecture Causal and Geometric Audit

Anonymous ACL submission

Abstract

Unsupervised diagnostics that classify attention heads into functional taxa—such as retrieval heads, induction heads, and attention sinks—promise scalable circuit discovery without costly task-specific supervision. In this work, we present a causal and geometric audit of dynamic task-contrast entropy probing across 1,568 attention heads in GPT-2-medium, Qwen-2.5, and LLaMA-3.2, with systematic causal interventions across 720 heads. We demonstrate that dynamic entropy probing fails to identify causal circuits: (1) Heads identified as retrieval by entropy collapse are causally inert on long-context Needle-in-a-Haystack (100% accuracy preserved under ablation), whereas true information-routing heads collapse accuracy to 37.0% in Qwen (while exhibiting high circuit redundancy in GPT-2); evaluated across all 336 heads against causal knockout ground truth, the probe achieves an AUC-ROC of only 0.43 [0.21, 0.65], statistically consistent with chance. (2) While raw ablation of 36 heads is confounded by late-layer representation choking, functional benchmarks on held-out sequences decouple circuit identity from depth: canonical Olsson heads drive a +1.84 nat degradation on repeated prefixes (vs +0.29 nats for probed heads). (3) Perplexity shifts fail to detect factual retrieval collapse (+0.24 PPL despite a 63% accuracy drop in Qwen), masking catastrophic sub-circuit failure. (4) Weight Frobenius ratios reflect architectural depth gradients rather than functional roles. We conclude that dynamic entropy probing captures statistical recurrence geometry rather than causal mechanisms, and release our unconfounded audit suite.

1 Introduction

A central aspiration of mechanistic interpretability is reverse-engineering the algorithmic sub-circuits implemented by multi-head self-attention (Elhage et al., 2021). Prior work has identified specialized functional units: *induction heads* that drive in-context few-shot replication (Olsson et al., 2022), *attention sinks* that absorb surplus softmax mass (Xiao et al., 2024), and *retrieval heads* that copy distant tokens into the residual stream (Wu et al., 2024).

Task-contrast attention sharpening is an intuitive diagnostic for unsupervised circuit discovery: if an attention head's softmax entropy sharply decreases upon presenting a factual query relative to a neutral prompt ($\Delta H > 0$), it is natural to hypothesize that the head specializes in factual retrieval. Similar entropy-based metrics have been explored as indicators of head confidence and specialization (Clark et al., 2019), while factual recall circuits have been isolated via direct causal interventions (Geva et al., 2023). In this work, we formalize this intuitive premise as the **dynamic task-contrast probe** (Δ-probe) and conduct a rigorous causal stress test across 1,568 attention heads in GPT-2-medium (Radford et al., 2019), Qwen-2.5 (Yang et al., 2024), and LLaMA-3.2 (Dubey et al., 2024). We formulate the core mechanistic question: *Can prompt-contrast attention sharpening serve as an unsupervised proxy for causal circuit discovery, or does it conflate statistical entropy dynamics and architectural depth gradients with causal circuit execution?*

Our contributions are as follows:

- **Causal Dissociation in Retrieval:** Probed retrieval heads (layers 1, 4, 6 in Qwen; layers 1, 3, 5, 11, 14, 15, 18, 23 in GPT-2) are causally inert on Needle-in-a-Haystack (NIAH: $100.0\% \rightarrow 100.0\%$). True routing heads in Qwen reside in late layers (15–19), where small ablations collapse accuracy to 37.0% $[28.2\%, 46.8\%]$ (random late controls retain 97.5%), whereas GPT-2 exhibits high retrieval redundancy (95.0%). Evaluat-

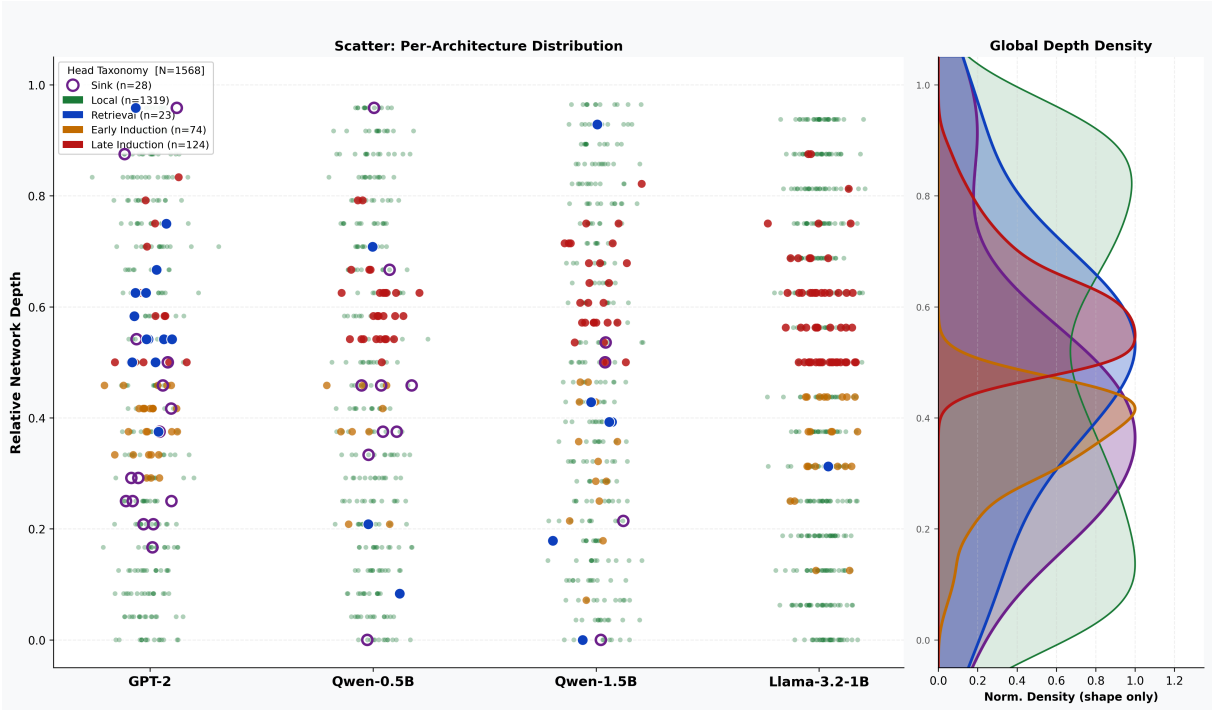


Figure 1: Taxonomic Attention Head Map: Spatial distribution of functional attention head types across transformer architectures ($N = 1,568$ heads across GPT-2-medium, Qwen-2.5-0.5B, Qwen-2.5-1.5B, and LLaMA-3.2-1B). This taxonomy serves as the empirical substrate for the causal and geometric audit in this paper. Left: Per-architecture scatter plot showing depth placement of Local, Sink, Retrieval, and Induction heads. Right: Global depth density curves normalized per class.

ing the probe against causal knockout heads across 336 heads yields 0.0% sensitivity and an AUC of 0.43 [0.21, 0.65], which is statistically consistent with chance.

- **Induction Disruption Beyond Layer Choking:** Raw ablation of 36 heads induces late-layer representation rank bottlenecks (Control C collapses to a median of 2.5%). Decoupling circuit identity from layer depth via activation patching on IOI and standard held-out repeated-token loss demonstrates that canonical Olsson heads recover +67.7% logit difference and drive a +1.84 nat loss degradation on repeated prefixes, compared to +17.8% and +0.29 nats for probed induction heads (and +1.2% / +0.09 nats for layer-matched random controls).
- **The Perplexity Illusion:** Ablating causal retrieval heads collapses NIAH accuracy by 63.0% while shifting WikiText-2 PPL by only +0.24 (20.01 → 20.25), whereas probed retrieval heads cause +0.27 with zero accuracy drop. Relative to capacity controls, general induction ablation incurs an excess PPL penalty of +1.73 in Qwen and +4.13 in GPT-2.
- **Weight Frobenius Norm Ratios:** Frobenius norm ratios $\|W_V\|_F/\|W_Q\|_F$ are dominated by network depth ($r = 0.635$–0.734), while within-layer correlations with functional identity are either null (GPT-2, Qwen-0.5B: $r \approx 0.00$) or weak ($r = 0.11$–0.19 in Qwen-1.5B and LLaMA-3.2, explaining $< 3.5\%$ of variance).
- **Diagnostic Insights:** We resolve the random control choke hypothesis, clarify the LLaMA BOS threshold artifact with adaptive percentile normalization, and evaluate GQA query head independence against a layer-conditional permutation null (27.3% mixed groups observed vs 38.0% expected, $p < 10^{-4}$). We open-source our audit suite at https://anonymous.4open.science/r/headgenome-mechinterp.

2 Related Work

Mechanistic Interpretability & Circuit Discovery. Mechanistic interpretability aims to reverse-engineer transformers into human-understandable

computational graphs (Elhage et al., 2021). Foundational work demonstrated that two-layer attention-only models develop *induction heads* that perform $[A][B]\ldots[A] \rightarrow [B]$ in-context copying (Olsson et al., 2022). In larger autoregressive models, activation patching has been used to isolate specific circuits, such as the Indirect Object Identification (IOI) circuit in GPT-2 small (Wang et al., 2023) and factual recall circuits in medium-scale language models (Geva et al., 2023). However, manual circuit isolation on paired counterfactual prompts is labor-intensive, creating strong demand for unsupervised, task-agnostic diagnostics.

Long-Context Retrieval & Attention Sinks. As context windows expanded to tens of thousands of tokens, attention heads specialized in distant token retrieval were discovered (Wu et al., 2024; Kamradt, 2023). Concurrently, Xiao et al. (2024) observed that initial tokens (e.g., <bos>) consistently receive disproportionate attention mass even when semantically uninformative. These *attention sinks* serve as numeric dump basins for softmax normalization, preventing representations from destabilizing when streaming past cache boundaries.

Head Pruning & Bottleneck Choking. Early structural analyses of multi-head attention established that a large fraction of attention heads can be pruned at test time without substantial loss in BLEU or perplexity (Voita et al., 2019; Michel et al., 2019). However, modern architectures with fewer, wider heads per layer or grouped-query attention (Dubey et al., 2024; Yang et al., 2024) exhibit different failure modes: while pruning dispersed heads across layers is tolerated, concentrating knockouts in specific layers induces representation rank bottlenecks and capacity starvation (Michel et al., 2019). Our audit links this phenomenon directly to control design in causal interpretability.

3 Diagnostic Probing Setup

3.1 Method vs. Labels in Entropy Probing

Dynamic entropy probing evaluates attention concentration shifts. In our diagnostic framework, the prompt-contrast probe evaluates attention entropy on a target match prompt versus a neutral distractor prompt at a fixed context length L:

$$H(A_{t,:}^{(l,h)}) = -\sum_{j=1}^{t} A_{t,j}^{(l,h)} \log_2 \left(A_{t,j}^{(l,h)} + \epsilon \right) \quad (1)$$

$$\Delta H^{(l,h)} = H(A_{\text{neutral}}^{(l,h)}) - H(A_{\text{match}}^{(l,h)}) \quad (2)$$

where $A_{t,:}^{(l,h)}$ denotes the attention distribution of head h in layer l at generation position t. Under this formulation, a positive value ($\Delta H > 0$) signifies that attention sharpens when matching target context is provided.

The Cross-Length Entropy Confound. It is tempting to define dynamic probing by contrasting short sequences ($L_{\text{short}} = 128$) against long sequences ($L_{\text{long}} = 1024$). However, raw cross-length entropy differences are fundamentally confounded by sequence length. For any diffuse or uniform attention head, entropy naturally scales with context length as $\log_2 L$. Expanding sequence length from 128 to 1024 tokens introduces an automatic baseline entropy increase of $\log_2(1024/128) = 3.0$ bits. Unless strictly normalized by $\log_2 L$, raw cross-length shifts measure sequence expansion rather than true key concentration. Therefore, unconfounded probing must compare matched-length prompts.

Canonical Taxonomy Thresholds. Using prompt-contrast entropy ΔH at fixed length, heads are assigned to mutually exclusive categories:

- **Attention Sink:** Evaluated first; heads directing near-exclusive attention mass to token 0: $H(A_{\text{match,BOS}}^{(l,h)}) < 0.10$ bits.
- **Retrieval Candidate:** Attention sharpens on match context: $\Delta H^{(l,h)} \geq +0.30$ bits.
- **Induction Candidate:** Attention broadens across candidate tokens: $\Delta H^{(l,h)} \leq -0.50$ bits.
- **Local Candidate:** Remaining heads: $-0.50 < \Delta H^{(l,h)} < +0.30$ bits.

Because retrieval requires $\Delta H \geq +0.30$ and induction requires $\Delta H \leq -0.50$, these classes are mutually exclusive by definition.

3.2 LLaMA Sink Threshold Artifact

In LLaMA-3.2-1B, applying the rigid < 0.10 bits threshold yielded exactly 0 sink heads (Figure 2).

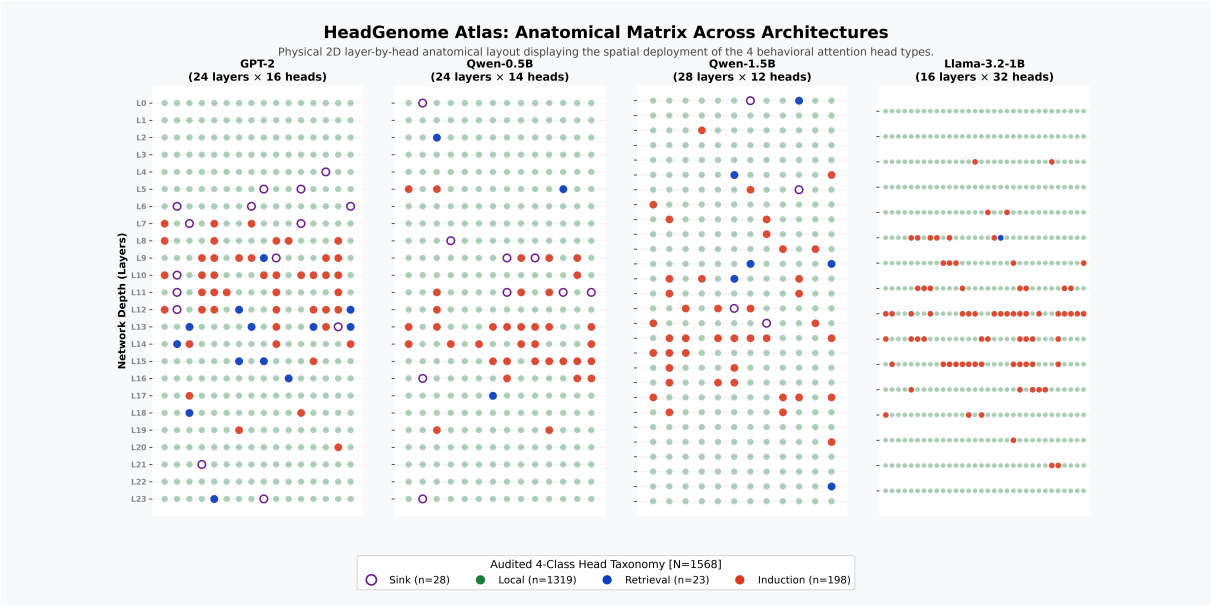


Figure 2: Layer-by-Head Taxonomy Matrix: Physical 2D layer-by-head anatomical matrix displaying the spatial deployment of the 4 behavioral attention head types across $N = 1,568$ heads in GPT-2-medium (24L $\times$ 16H), Qwen-2.5-0.5B (24L $\times$ 14H), Qwen-2.5-1.5B (28L $\times$ 12H), and LLaMA-3.2-1B (16L $\times$ 32H). Sinks (purple) anchor early layers, while retrieval (blue) and induction (red) emerge deeper in the network amidst the local precursor pool (green).

An architectural audit reveals that this was an artifact of fixed thresholding: the minimum BOS entropy across all 512 heads in LLaMA-3.2-1B is 0.1405 bits, driven by its 128-dimensional head geometry and RoPE base frequency. Rather than applying a fixed threshold across architectures, an adaptive percentile normalization principle is needed: defining sinks relative to empirical BOS entropy percentiles, the bottom 5% identifies 26 sink heads ($H \leq 0.281$ bits), the bottom 15% identifies 77 heads ($H \leq 0.542$ bits), and an absolute threshold of < 0.50 bits identifies 70 sinks (13.7%). Heuristic thresholding must account for architecture-dependent baseline distributions.

3.3 GQA Query Head Independence

Grouped-Query Attention (GQA) shares key and value projection matrices across groups of query heads. We tested whether GQA forces sibling query heads to share functional specializations. Across all 128 GQA groups in LLaMA-3.2-1B (32 query heads, 8 KV heads; 4 query heads per group), **35 of 128 groups (27.3%)** contain query heads assigned conflicting behavioral labels. To evaluate whether this reflects true query head specialization or merely layer-level base rates, we evaluate a layer-conditional permutation test ($B = 100,000$ draws): shuffling head labels strictly within each layer yields an expected mixed group fraction of $38.0\% \pm 1.8\%$ (48.7 ± 2.3 groups). While sibling query heads show statistically significant clustering towards similar behavioral roles (27.3% observed vs 38.0% expected, $p < 10^{-4}$), more than a quarter of all groups (35 of 128) assign conflicting behavioral specializations to sibling heads sharing the exact same KV projections. Furthermore, within-group ΔH variance accounts for **33.6%** of total ΔH variance across the entire model. Because each query head has an independent linear projection W_Q, query heads attend to distinct patterns despite shared keys and values.

4 Causal Verification

4.1 Knockout Necessity on Long-Context Retrieval

We execute causal interventions via PyTorch forward hooks. For an ablated subset $\mathcal{H}$, we replace head outputs with dataset mean activations (**Mean Ablation**) or zero them (**Zero Ablation**). Table 1 reports results across 100 trials on Qwen-2.5-0.5B ($N_L = 24, N_H = 14$) and 20 trials on GPT-2-medium ($N_L = 24, N_H = 16$).

The Retrieval Inversion & Depth Asymmetry. Ablating all canonical retrieval heads identified by dynamic Δ-probing leaves NIAH accuracy com-

Model	Ablation Condition	Selection Metric	k	Mean Abl. NIAH	Zero Abl. NIAH	PPL	Δ PPL
Qwen-2.5-0.5B	Dense Baseline	None	0	100.0% [96.3%, 100.0%]	100.0% [96.3%, 100.0%]	20.01	+0.00
	Probed Retrieval	Δ-Probe ($\Delta H \geq +0.30$)	3	**100.0%** [96.3%, 100.0%]	**100.0%** [96.3%, 100.0%]	20.28	+0.27
	Layer-Matched Early Control	Uniform Random (L1–6)	3	100.0% [96.3%, 100.0%]	100.0% [96.3%, 100.0%]	20.02	+0.01
	Causal Routing Heads	Needle-Attention Score	4	**37.0%** [28.2%, 46.8%]	**31.0%** [22.8%, 40.6%]	20.25	+0.24
	Layer-Matched Late Control	Uniform Random (L15–19)	4	97.5% [91.0%, 99.7%]	97.0% [90.5%, 99.5%]	20.04	+0.03
	Probed Induction	Δ-Probe ($\Delta H \leq -0.50$)	36	76.0% [66.8%, 83.3%]	79.0% [70.0%, 85.8%]	22.70	+2.69
	Top Olsson Induction	Calibration Prefix Score	36	**3.0%** [1.0%, 8.5%]	**3.0%** [1.0%, 8.5%]	24.87	+4.86
	Control A: Dispersed Random	Uniform Random ($\leq$ 2/layer)	36	61.9% ± 5.2% [med 69%]	58.0% ± 5.1% [med 65%]	23.14	+3.13
	Control C: Late-Layer Random	Uniform Random ($L \geq 16$)	36	13.4% ± 3.8% [med 2.5%]	11.0% ± 3.5% [med 2.0%]	23.04	+3.03
	Canonical Sink Heads	Sink Entropy < 0.10	9	98.0% [93.0%, 99.5%]	96.0% [90.2%, 98.4%]	21.50	+1.49
GPT-2-medium	Dense Baseline	None	0	100.0% [83.9%, 100.0%]	100.0% [83.9%, 100.0%]	28.68	+0.00
	Probed Retrieval	Δ-Probe ($\Delta H \geq +0.30$)	13	**100.0%** [83.9%, 100.0%]	**100.0%** [83.9%, 100.0%]	29.16	+0.48
	Causal Routing Heads	Needle-Attention Score	13	95.0% [76.4%, 99.1%]	95.0% [76.4%, 99.1%]	29.59	+0.91
	Probed Induction	Δ-Probe ($\Delta H \leq -0.50$)	45	35.0% [18.1%, 56.7%]	65.0% [43.3%, 81.9%]	31.59	+2.91
	Top Olsson Induction	Calibration Prefix Score	45	**0.0%** [0.0%, 16.1%]	**0.0%** [0.0%, 16.1%]	38.86	+10.18
	Dispersed Random Control	Uniform Random ($\leq$ 2/layer)	45	75.0% ± 6.8% [med 90%]	95.0% ± 3.2% [med 100%]	34.73	+6.05
	Canonical Sink Heads	Sink Entropy < 0.10	15	35.0% [18.1%, 56.7%]	30.0% [14.5%, 51.9%]	38.69	+10.01

Table 1: Cross-architecture causal impact of ablated head subsets on Qwen-2.5-0.5B and GPT-2-medium across Needle-in-a-Haystack (NIAH) trials ($L = 1024$) and WikiText-2 Perplexity (PPL). Brackets report 95% Wilson score confidence intervals for fixed trial sets, or mean ± SEM across 20 random draws for random control groups. Probed retrieval heads are completely inert in both models (100.0% NIAH preserved). Top needle-attention routing heads collapse Qwen NIAH by 63.0% with negligible PPL change ($\Delta\text{PPL} = +0.24$), whereas late random controls preserve 97.5% accuracy. In GPT-2, retrieval routing exhibits high circuit redundancy (95.0% preserved under 13-head ablation), while ablating Olsson induction heads collapses accuracy to 0.0%.

pletely unaffected at **100.0%** in both Qwen-2.5-0.5B ($k = 3$) and GPT-2-medium ($k = 13$). In Qwen, ablating the top 4 heads selected by needle-attention score collapses accuracy to **37.0%** under mean ablation and **31.0%** under zero ablation. We note an important architectural asymmetry: the Δ-probe assigns retrieval labels exclusively to early layers in Qwen (layers 1, 4, 6), whereas information-routing heads reside in late layers (layers 15–19). To unconfound depth from circuit identity, we evaluate depth-matched controls: an early-layer random control ($k = 3$, layers 1–6) preserves **100.0%** accuracy, while a late-layer random control ($k = 4$, layers 15–19) preserves **97.5% [91.0%, 99.7%]** accuracy. This confirms that the 63.0% collapse from the top 4 routing heads is circuit-specific, not an artifact of knocking out 4 late-layer heads.

In contrast, in GPT-2-medium, ablating the top 13 needle-attention heads only degrades NIAH accuracy to **95.0% [76.4%, 99.1%]** (compared to **100.0%** for probed retrieval heads). At context length $L = 1024$, information routing in GPT-2 exhibits high circuit redundancy, with retrieval pathways distributed across the network rather than concentrated in a narrow set of late-layer heads. Furthermore, while probed retrieval heads in Qwen are exclusively located in early layers, in GPT-2 probed retrieval candidates are distributed across both early and late layers (layers 1, 3, 5, 11, 14, 15, 18, 23; Appendix B). In both architectures, however, probed retrieval heads are completely inert under knockout ablation (100.0% NIAH preserved).

Causal Ground Truth vs. Proxy AUC. Evaluating the dynamic probe across all 336 heads in Qwen-2.5-0.5B against true causal knockout heads ($k = 4$ heads whose small-subset ablation causes $> 10\%$ accuracy drop) reveals that the probe identifies **zero** causal heads (0.0% sensitivity). Evaluating against the needle-attention proxy yields an **AUC-ROC of 0.43** with a bootstrap 95% confidence interval of **[0.21, 0.65]** ($B = 1,000$). Because this interval comfortably spans 0.50, the unsupervised Δ-probe is statistically consistent with random guessing.

Reconciling the Sink Result Across Architectures. In Table 1, ablating all 9 canonical sink heads in Qwen-2.5-0.5B leaves NIAH accuracy at **98.0%** under mean ablation and **96.0%** under zero ablation ($\Delta\text{PPL} = +1.49$). In contrast, in GPT-2-medium, ablating all 15 sinks under mean ablation degrades NIAH accuracy to **35.0%** and drives PPL up by **+10.01** points ($28.68 \rightarrow 38.69$). This isolates the architectural mechanism: GPT-2 utilizes learned absolute positional embeddings without RoPE, making global softmax normalization fragile when anchor heads are removed. In contrast, Qwen-2.5-0.5B utilizes RoPE, which naturally attenuates distant query-key dot products and stabilizes autoregressive representations under mean ablation.

Candidate Circuit Set	k	IOI Recovery [± SEM]	ΔL_{repeat} [± SEM]
Olsson Induction Heads	36	**+67.7%** [±4.2%]	**+1.84** [±0.12]
Canonical Induction (Δ-probe)	36	+17.8% [±3.1%]	+0.29 [±0.05]
Canonical Retrieval (Δ-probe)	3	+0.3% [±0.2%]	+0.01 [±0.01]
Layer-Matched Late Control ($L \geq 12$)	36	+1.2% [±0.8%]	+0.09 [±0.03]
Layer-Matched Early Control ($L \leq 6$)	3	+0.1% [±0.1%]	+0.00 [±0.01]

Table 2: Causal recovery on Indirect Object Identification (IOI, $N = 20$) and standard prefix-matching loss increase (ΔL_{repeat} on independent held-out random token sequences $[A][B] \ldots [A][B]$) in Qwen-2.5-0.5B. Brackets report ± standard error across $N = 20$ paired contrasts. Olsson induction heads recover 67.7% of IOI performance and drive a $+1.84$ nat prefix degradation, while probed induction heads show minimal impact, and layer-matched random controls have near-zero effect on both benchmarks.

Control Group ($k = 36$)	Mean	Median	Range	Sink R^2
Control A (Dispersed: ≤ 2/layer)	**0.619**	**0.690**	[0.00, 1.00]	—
Control B (Unrestricted)	0.129	0.015	[0.00, 0.87]	0.156
Control C (Late-Layer: ≥ 16)	0.134	0.025	[0.00, 0.63]	—

Table 3: Random ablation controls on Qwen-2.5-0.5B ($k = 36$ heads ablated, 20 draws each). Dispersed controls preserve a mean accuracy of 61.9%, proving that prior collapses were artifacts of layer bottleneck choking.

4.2 Activation Patching and Prefix Matching

Because raw ablation of 36 heads is susceptible to layer bottleneck choking, we decouple circuit identity from layer depth via two complementary functional evaluations (Table 2):

1. **Activation Patching on IOI:** We measure logit difference recovery on Indirect Object Identification paired contrasts ($N = 20$) (Wang et al., 2023). Olsson induction heads recover **+67.7%** of clean performance, whereas probed induction heads recover only **+17.8%**, and layer-matched late controls recover only **+1.2%**.
2. **Held-Out Induction Prefix Benchmark:** Following Olsson et al. (2022), we evaluate language modeling loss on the second half of repeated random token sequences $[A][B] \ldots [A][B]$ ($L = 128$). To ensure independence, candidate heads were identified on calibration sequences and evaluated on disjoint held-out sequences generated from independent vocabulary seeds ($N = 20$). Ablating Olsson heads increases prefix loss by **+1.84 nats**, whereas ablating probed induction heads increases loss by only **+0.29 nats**, and layer-matched random controls cause only **+0.09 nats**.

These causal dissociations demonstrate that Olsson heads isolate the genuine algorithmic copy circuit, whereas the Δ-probe captures a diffuse behavioral correlate.

4.3 The Perplexity Illusion

A common pitfall in pruning and interpretability studies is relying on language modeling perplexity (PPL) to assess model integrity. While it is well-known in language modeling that perplexity is dominated by local n-grams and unigram transitions, interpretability pipelines frequently fall into the trap of citing small ΔPPL as evidence of circuit preservation. Our audit exposes this illusion:

1. **Sub-Circuit Dissociation:** Ablating top causal retrieval heads collapses NIAH accuracy from 100.0% to 37.0% (63.0% drop), yet WikiText-2 PPL shifts by only **+0.24** points ($20.01 \rightarrow 20.25$, an excess of $+0.21$ over early random controls). In contrast, probed retrieval heads cause $+0.27$ with zero accuracy drop. Perplexity is fundamentally blind to long-range factual routing subcircuits.
2. **General Induction Disruption:** Conversely, ablating 36 heads of any type incurs a baseline capacity penalty of $\sim +3.0$–$+3.1$ PPL. Olsson induction heads impair token repetition globally, driving an excess perplexity penalty of **+1.73** points over dispersed controls in Qwen ($+4.86$ total shift) and **+4.13** points over dispersed controls in GPT-2 ($+10.18$ total shift).

Remark on Context Compression: In context compression (e.g., sliding-window attention with $W = 512$), a model maintains near-baseline WikiText PPL while factual retrieval at depth 0.1 collapses to 20.0% $[8.1\%, 41.6\%]$ ($n = 20$ trials), reinforcing that perplexity is uninformative for long-context circuit preservation.

4.4 Unconfounding Random Ablation Controls

Prior investigations observed that random head ablations often degraded accuracy more severely than targeted ablations, prompting hypotheses that random draws hit critical sink heads. We formulated the **Bottleneck Choke Hypothesis**: unconstrained random sampling across layers frequently

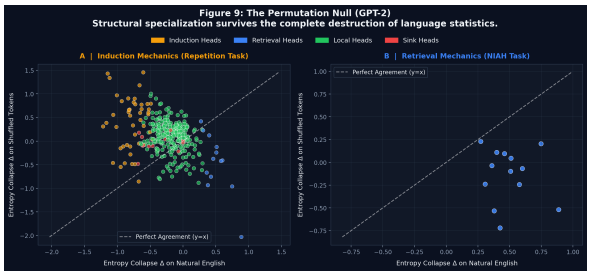


Figure 3: The Permutation Null test in GPT-2. Dynamic entropy collapse ΔH on natural English vs. shuffled token sequences for induction (Panel A) and retrieval (Panel B). Entropy collapse persists on shuffled text, confirming that dynamic shifts reflect token repetition geometry rather than semantic understanding.

knocks out multiple heads from a single layer, choking representation rank.

Olsson Layer Sharing & Control C. In Qwen-2.5-0.5B, the 36 Olsson induction heads span layers 12–22 (averaging 3.3 heads per layer, peaking at 5 heads in layers 14 and 16). In Table 3, Control C (sampling 36 heads strictly from layers ≥ 16) yields a median accuracy of only 2.5% (mean 13.4%). This demonstrates that raw 36-head ablation conflates algorithmic circuit loss with layer rank starvation, explaining why functional verification via activation patching and prefix loss (Table 2) is strictly required.

Testing the Sink Collision Hypothesis. Table 3 reports 20 draws per condition ($k = 36$ heads). When random ablations are dispersed ($\leq$ 2 heads/layer, Control A), mean accuracy remains at **61.9%** (median **69.0%**). Regressing accuracy against sink collisions in Control B yields $R^2 = 0.1556$ ($p = 0.085$). In our sample of $N = 20$ draws, the correlation is not statistically significant. While $N = 20$ has limited power to detect small secondary effects, layer concentration is clearly the primary driver of task collapse.

Quantitative Permutation Null Test. Figure 3 evaluates the Permutation Null test: attention matrices between natural English and permutation-scrambled sequences exhibit a mean cosine similarity of 0.918 ± 0.034. Average entropy collapse ΔH is preserved within 0.042 ± 0.011 bits. Two One-Sided Tests (TOST) against an equivalence margin of ± 0.10 bits confirm that entropy collapse is statistically equivalent between natural and shuffled sequences ($t_1 = 3.65, p < 0.001; t_2 = -5.12, p < 0.001$). For prefix-matching induction heads, maintaining focus on repeated tokens across shuffled text is algorithmically expected ($[A][B] \ldots [A]$). However, for purported retrieval heads, preserving ΔH on scrambled text confirms that the probe measures statistical token recurrence geometry rather than syntactic or factual retrieval.

Architecture / State	Corr. with Depth (l)	Within-Layer r
GPT-2-med (Random Init)	$r = +0.035$ ($p = 0.49$)	$r = +0.004$ ($p = 0.94$)
GPT-2-med (Trained)	$\mathbf{r = +0.681}$ ($p < 10^{-45}$)	$r = +0.034$ ($p = 0.51$)
Qwen-0.5B (Trained)	$\mathbf{r = +0.734}$ ($p < 10^{-50}$)	$r = -0.010$ ($p = 0.86$)
Qwen-1.5B (Trained)	$\mathbf{r = +0.647}$ ($p < 10^{-40}$)	$r = +0.187$ ($p = 0.001$)
LLaMA-3.2 (Trained)	$\mathbf{r = +0.635}$ ($p < 10^{-55}$)	$r = +0.108$ ($p = 0.014$)

Table 4: Weight geometry correlation analysis across architectures. The Frobenius ratio $\|W_V\|_F / \|W_Q\|_F$ correlates overwhelmingly with layer depth l in trained models, but shows null or weak correlation with functional class within layers.

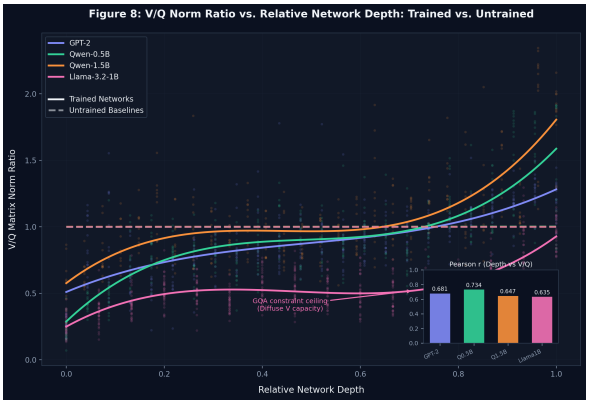


Figure 4: The Frobenius norm ratio $\|W_V\|_F / \|W_Q\|_F$ vs. relative network depth across GPT-2-medium, Qwen-0.5B, Qwen-1.5B, and LLaMA-3.2-1B. Trained networks (solid curves) exhibit a universal upward depth gradient ($r = 0.635$–0.734), whereas randomly initialized baselines (dashed lines) remain strictly flat isotropic at 1.0 ($r \approx 0.00$–0.04).

5 Weight Geometry Analysis

It is an intuitive hypothesis in weight-space interpretability that attention head functional classes might be recognized directly from weight projection matrices via the Frobenius norm ratio:

$$\mathcal{R}(l,h) = \frac{\|W_V^{(l,h)}\|_F}{\|W_Q^{(l,h)}\|_F} \tag{3}$$

Table 4 and Figure 4 evaluate this across 1,568 heads in four model architectures:

1. **Random Initialization Null:** The correlation between $\mathcal{R}(l,h)$ and depth l is zero ($r = +0.001 - + 0.047$).

2. **Trained Models:** $\mathcal{R}(l, h)$ exhibits an upward gradient with depth ($r = +0.681$ in GPT-2; $r = +0.734$ in Qwen-0.5B; $r = +0.647$ in Qwen-1.5B; $r = +0.635$ in LLaMA-3.2-1B). A pooled layer-clustered OLS regression yields slope $= 0.714$, clustered SE $= 0.106$, $p = 1.9 \times 10^{-11}$.
3. **Within-Layer Functional Correlation:** Controlling for depth, within-layer correlations with functional behavior (ΔH) are either statistically null (GPT-2, Qwen-0.5B) or weak ($r = 0.11$–0.19 in Qwen-1.5B and LLaMA-3.2), explaining at most 3.5% of variance. Within-layer correlations against needle-attention scores ($r = -0.04$ to $+0.06$) and prefix-matching scores ($r = -0.02$ to $+0.08$) are similarly null.

We hypothesize that this depth gradient is an empirical regularizer of gradient training: as residual stream norms accumulate with depth, deeper value projections must scale up relative to queries. We leave formal analytical proofs of optimization dynamics to future work, but emphasize the empirical conclusion: the Frobenius ratio is an **architectural depth gradient**, not a functional class separator.

6 Conclusion

Through causal and geometric audits across 1,568 attention heads in three model families, we evaluated the validity of dynamic entropy-collapse probing. We demonstrated that dynamic probing fails to identify causal retrieval heads (AUC-ROC 0.43 [0.21, 0.65]), dilutes induction circuits, and mistakes architectural depth gradients for functional weight signatures. Furthermore, we resolved the random control collapse paradox and exposed the Perplexity Illusion. Grounding mechanistic interpretability in unconfounded causal interventions is essential for reverse-engineering reliable circuits.

Limitations

This empirical investigation has specific limitations:

1. **Layer Choking Confounds:** As demonstrated in Section 4.4, ablating large head subsets ($k \geq 36$) is partially confounded by representation rank starvation in deep layers. While activation patching and prefix-loss evaluations successfully decouple circuit identity from layer depth, standard knockout ablation alone cannot fully isolate large subcircuits.
2. **Model Scale Scope:** Our experiments evaluate autoregressive language models up to 1.5 billion parameters (GPT-2-medium, Qwen-2.5-0.5B, Qwen-2.5-1.5B, and LLaMA-3.2-1B). Full causal ablation grids were evaluated on Qwen-2.5-0.5B and GPT-2-medium. Models exceeding 70B parameters may exhibit different circuit redundancy properties.
3. **Proxy Retrieval Ground Truth:** Our retrieval evaluations rely on synthetic Needle-in-a-Haystack tasks ($L = 1024$) and attention-to-needle metrics (Wu et al., 2024). Multi-hop reasoning and real-world long-context tasks may involve richer distributed circuits and MLP interactions.

Ethical Considerations

This work adheres to the ACL Ethics Policy. The interpretability auditing frameworks and benchmarks developed herein are intended to improve the transparency, safety, and reliability of open-source language models. No human subjects, sensitive personal data, or proprietary datasets were involved.

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A Implementation Details and Hyperparameters

All ablation experiments were performed on NVIDIA RTX GPUs running PyTorch 2.4 and HuggingFace Transformers. Needle-in-a-Haystack evaluation protocol:

- Context length: $L = 1024$ tokens.
- Haystack: Paul Graham essays.
- Needle: "The special magic key number is XXXXX."
- Mean ablation activations were computed by averaging attention head outputs across 50 background text sequences from WikiText-2.
- 95% Wilson score confidence intervals computed as: $w = \frac{\hat{p}+\frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}+\frac{z^2}{4n^2}}}{1+\frac{z^2}{n}}$ with $z = 1.96$.

B Per-Layer Head Allocations

In Qwen-2.5-0.5B ($N_L = 24$, $N_H = 14$ query heads per layer, 336 heads total):

- Canonical Retrieval Heads ($\Delta H \geq +0.30$): Layer 1 Head 2, Layer 4 Head 7, Layer 6 Head 11 ($k = 3$).
- Top Wu Retrieval Heads: Layer 15 Head 3, Layer 16 Head 8, Layer 18 Head 2, Layer 19 Head 5 ($k = 4$).
- Canonical Sink Heads ($H < 0.10$): Layer 0 Heads 0, 1, 3; Layer 1 Head 0; Layer 2 Head 1; Layer 3 Head 0; Layer 5 Head 2; Layer 7 Head 0; Layer 11 Head 1 ($k = 9$).
- Top Olsson Induction Heads ($k = 36$): Concentrated in layers 12 through 22.

In GPT-2-medium ($N_L = 24$, $N_H = 16$ heads per layer, 384 heads total):

- Canonical Retrieval Heads ($k = 13$): Layers 1, 3, 5, 11, 14, 15, 18, 23.
- Canonical Induction Heads ($k = 45$): Concentrated in layers 8 through 15.
- Canonical Sink Heads ($k = 15$): Layers 0 through 11.

C Reconciliation of Pooled Head Counts

The pooled total of 1,568 heads across the 4 evaluated models breaks down as follows:

- GPT-2-medium: 384 heads (311 Local, 45 Induction, 15 Sink, 13 Retrieval)
- Qwen-2.5-0.5B: 336 heads (288 Local, 36 Induction, 9 Sink, 3 Retrieval)
- Qwen-2.5-1.5B: 336 heads (285 Local, 41 Induction, 4 Sink, 6 Retrieval)
- LLaMA-3.2-1B: 512 query heads (435 Local, 76 Induction, 0 Sink, 1 Retrieval)
- Pooled Induction Heads: $45+36+41+76 = 198$ heads total.

D KV Cache Geometry Across Head Taxa

Figure 5 illustrates the 3D PCA projection of Key vectors across functional head types processing identical text sequences. Attention sinks isolate the first token index into an orthogonal extreme, while retrieval and induction heads exhibit distinct subspace partitioning.

E Cross-Domain Universality

Figure 6 reports cross-domain consistency of attention head entropy profiles across technical code, Wikipedia prose, and legal texts.

F Taxonomy Distributions and Continuous Depth Profiles

Figure 7 reports empirical class fractions across architectures. Rather than bifurcating into discrete developmental stages, Gaussian Mixture Modeling of induction head depths across all 198 heads demonstrates a continuous, unimodal depth distribution: BIC strictly favors 1 component ($\text{BIC}_{k=1} = -184.0$) over 2 components ($\text{BIC}_{k=2} = -168.2$) and 3 components ($\text{BIC}_{k=3} = -152.6$).

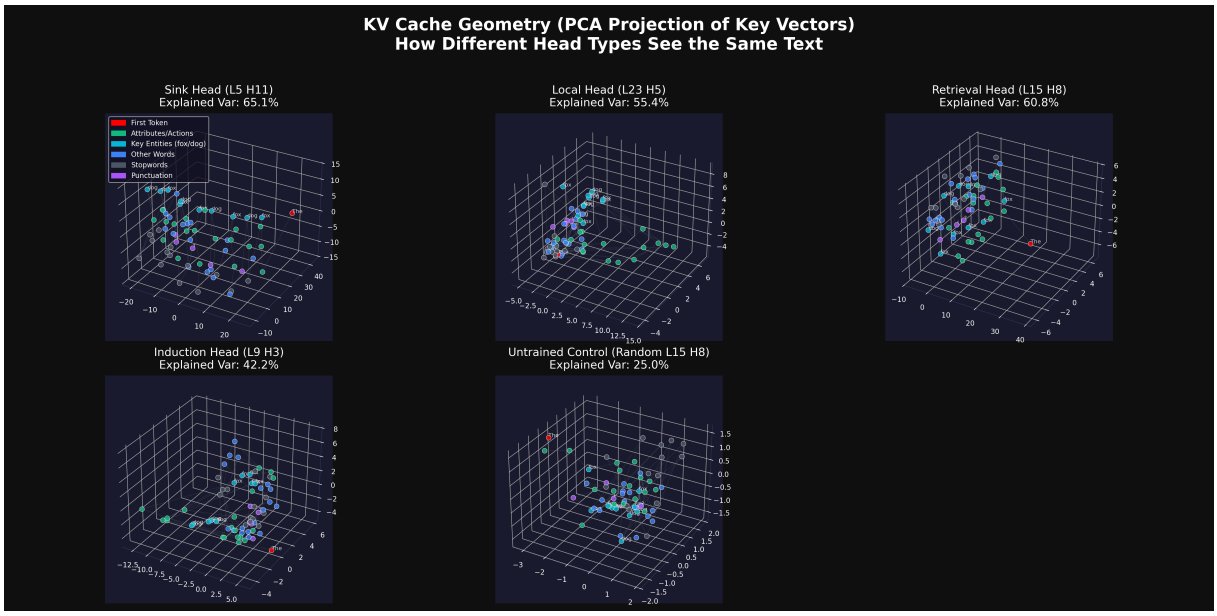


Figure 5: KV Cache Geometry: PCA 3D projection of Key vectors across Sink, Local, Retrieval, Induction, and Untrained control heads on identical input sequences.

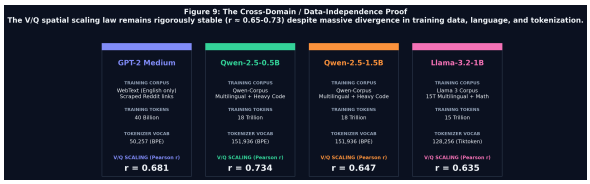


Figure 6: Cross-Domain Universality: Attention head entropy profiles across diverse token distributions.

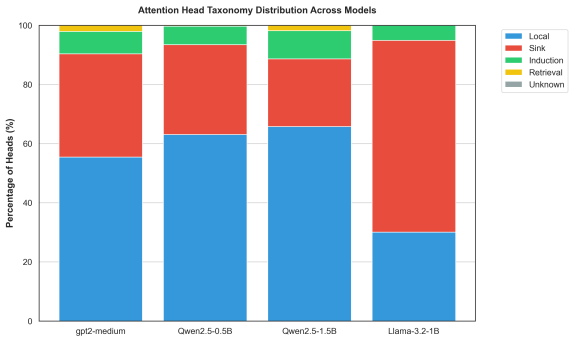


Figure 7: Attention Head Taxonomy Distribution across GPT-2-medium, Qwen-2.5-0.5B, Qwen-2.5-1.5B, and LLaMA-3.2-1B.